

Linux Fundamentals

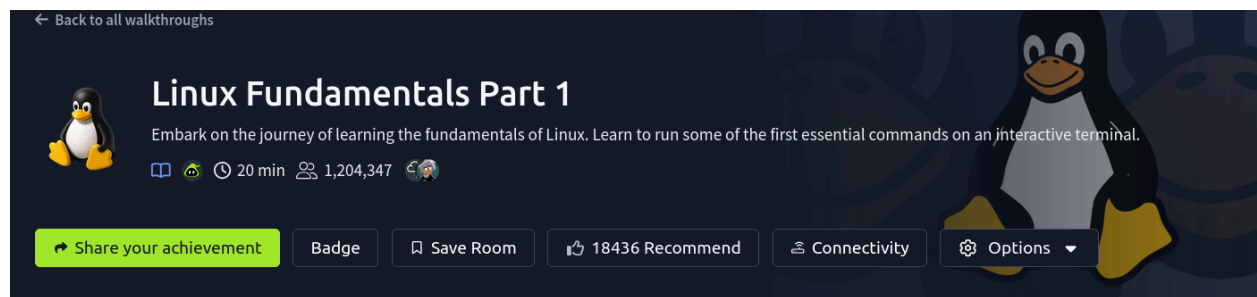
Platform:TryHackMe

Learning Path : Linux Fundamentals

Completion Status:Completed

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CTC-153-26



is actually an umbrella term for multiple 's that are based on UNIX (another operating system

Interacting With the Filesystem

As I previously stated, being able to navigate the machine that you are logged into without relying on a desktop environment is pretty important. After all, what's the point of logging in if we can't go anywhere?

Command	Full Name
ls	listing
cd	change directory
cat	concatenate
pwd	print working directory

Answer the questions below

On the Linux machine that you deploy, how many folders are there?

✓ Correct Answer

Which directory contains a file?

✓ Correct Answer



What is the contents of this file?

✓ Correct Answer

Use the cd command to navigate to this file and find out the new current working directory. What is the path?

✓ Correct Answer

searching files with find command and grep

Task 7 An Introduction to Shell Operators

Linux operators are a fantastic way to power up your knowledge of working with Linux. There are a few important operators that are worth noting. We'll cover the basics and break them down accordingly to bite-sized chunks.

At an overview, I'm going to be showcasing the following operators:

Symbol / Operator	Description
&	This operator allows you to run commands in the background of your terminal.
&&	This operator allows you to combine multiple commands together in one line of your terminal.
>	This operator is a redirector - meaning that we can take the output from a command (such as using cat to output a file) and direct it elsewhere.
>>	This operator does the same function of the > operator but appends the output rather than replacing (meaning nothing is overwritten).

Let's cover these in a bit more detail.

Answer the questions below

If we wanted to run a command in the background, what operator would we want to use?

`&` ✓ Correct Answer

If I wanted to replace the contents of a file named "passwords" with the word "password123", what would my command be?

`echo password123 > passwords` ✓ Correct Answer ?

Now if I wanted to add "tryhackme" to this file named "passwords" but also keep "passwords123", what would my command be?

`echo tryhackme >> passwords` ✓ Correct Answer ?

Now use the deployed Linux machine to put these into practice

No answer needed ✓ Correct Answer

Linux Fundamentals Part 2

The second part focused on interacting with remote Linux systems, searching for files, understanding permissions, and becoming more productive with command-line utilities.

One of the first concepts introduced was Secure Shell (SSH). SSH enables users to securely connect to remote Linux machines over a network using encrypted communication. Since many cybersecurity labs and cloud servers are accessed through SSH, learning this protocol was extremely valuable. I practiced connecting to remote systems, executing commands, and transferring files securely.

The module also introduced Linux permissions, which determine who can read, write, or execute files. Every file and directory has permissions assigned to three categories: the owner, the group, and others. Initially, the permission notation looked confusing, but after understanding the meaning of r, w, and x, it became much easier to interpret.

I learned how to modify permissions using the `chmod` command and change file ownership with `chown`. These commands are essential for system administration and also play an important role in privilege escalation during penetration testing.

Searching for files became significantly easier after learning commands such as `find` and `locate`. Instead of manually browsing directories, I could quickly identify files based on their names, types, or permissions.

Another useful lesson covered shell operators and command chaining. Using operators like `&&`, `||`, `>`, `>>`, and `|` allowed me to combine multiple commands, redirect output, and create

efficient command pipelines. This introduced me to the Unix philosophy of building complex operations by combining simple tools.

Task 1	✓	Introduction	▼
Task 2	✓	Accessing Your Linux Machine Using SSH (Deploy)	☰ ▼
Task 3	✓	Introduction to Flags and Switches	▼
Task 4	✓	Filesystem Interaction Continued	▼
Task 5	✓	Permissions 101	▼
Task 6	✓	Common Directories	▼
Task 7	✓	Conclusions and Summaries	▼
Task 8	✓	Linux Fundamentals Part 3	▼

Linux Fundamentals Part 3

The final part introduced more advanced Linux utilities that are commonly used in daily system administration and cybersecurity.

One of the most valuable topics was working with system processes. Linux constantly runs multiple processes simultaneously, and learning how to monitor and manage them is essential. I learned to view running processes using commands such as `ps` and `top`, terminate unwanted processes using `kill`, and understand process identifiers (PIDs).

The module also introduced package management. Installing, updating, and removing software is much simpler using package managers. Since TryHackMe primarily uses Ubuntu-based systems, I practiced using `apt` to install new software, update package lists, and upgrade installed applications.

Environment variables were another useful concept. Variables such as `PATH` determine where Linux searches for executable programs. Understanding environment variables helped me understand why commands can be executed from almost anywhere in the terminal.

The course also explained automation through shell scripting. Although only introductory concepts were covered, I learned that Bash scripts can automate repetitive tasks, making system administration and security testing much more efficient.

Another practical topic involved managing system logs. Linux records valuable information about user activity, services, and system events. Learning where these logs are stored and how to examine them provided useful skills for troubleshooting and forensic investigations.

Finally, the module introduced several everyday Linux utilities that improve productivity. Commands such as **grep**, **sort**, **wc**, **cut**, and **awk** demonstrated how powerful text processing can be in Linux. Since cybersecurity professionals frequently analyze log files, configuration files, and command outputs, these tools are extremely valuable.

Task 1	✔	Introduction	▼
Task 2	✔	Accessing Your Linux Machine Using SSH (Deploy)	☰ ▼
Task 3	✔	Introduction to Flags and Switches	▼
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Task 8	✔	Linux Fundamentals Part 3	▼